

# MECHATRONIKA - 5. gyakorlat

## Veérték tételek

$$f(0) = \lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} s \cdot F(s)$$

- kezleti érték tétele

$$f(\infty) = \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s \cdot F(s)$$

- végérték / állandósult állapot tétele

de! vigyáznunk kell, hogy  
van-e állandósult állapot

$$(1) F_1(s) = 3 \frac{s+2}{(s+1)(s+4)} \rightarrow f_1(t) = (e^{-t} + 2e^{-4t}) \varepsilon(t)$$

→ korábbi feladatokból tudjuk

$$\begin{aligned} \text{kezleti érték: } f_1(0) &= \lim_{s \rightarrow \infty} s \cdot 3 \frac{s+2}{(s+1)(s+4)} = \lim_{s \rightarrow \infty} 3 \frac{s^2+2s}{s^2+5s+4} = \\ &= \lim_{s \rightarrow \infty} 3 \frac{1+2/s}{1+5/s+4/s^2} = 3 \cdot 1 = \boxed{3} \end{aligned}$$

$$\text{állandósult állapot: } f_1(\infty) = \lim_{s \rightarrow 0} 3 \frac{s^2+2s}{(s+1)(s+4)} = 3 \cdot \frac{0}{4} = \boxed{0}$$

$$(2) F_2(s) = \frac{5s-12}{s^2-4s} = \frac{5s-12}{s(s-4)}$$

$$\text{kezleti érték: } f_2(0) = \lim_{s \rightarrow \infty} s \cdot \frac{5s-12}{s(s-4)} = \lim_{s \rightarrow \infty} \frac{5s-12}{s-4} = \lim_{s \rightarrow \infty} \frac{5-12/s}{1-4/s} = \boxed{5}$$

$$\text{állandósult állapot: } f_2(\infty) = \lim_{s \rightarrow 0} \frac{5s-12}{s-4} = \frac{-12}{-4} = \boxed{3}$$

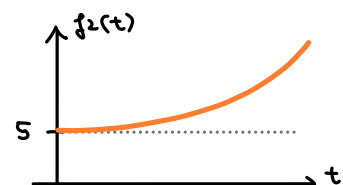
de! van állandósult állapot?

$$F_2(s) = \frac{5s-12}{s(s-4)} = \frac{A}{s} + \frac{B}{s-4}$$

$\swarrow$   $s^{-1}$   $\searrow$   $s^{-1}$   
 $A \cdot \varepsilon(t)$   $B \cdot e^{4t} \varepsilon(t)$  !

eszae  $e^{4t}$  miatt

meg kell vizsgálnunk, hogy  
az adott fv.-nek van-e  
állandósult állapota



## Differenciálegyenletek megoldása Laplace transzformációval

$$\textcircled{1} \dot{x}(t) = -3x(t) + 7u(t)$$

$$x(0) = 10$$

$u(t) = \varepsilon(t)$  gerjesztésre keressük a differenciálegyenlet  $x(t)$  megoldását

↓ Laplace

$$\mathcal{L}\{\dot{x}(t)\} = sX(s) - x(0)$$

$$sX(s) - x(0) = -3X(s) + 7U(s)$$

$$X(s)[s+3] = 7U(s) + x(0)$$

$$X(s) = \underbrace{\frac{7}{s+3}}_{\text{átviteli fv.}} U(s) + \underbrace{\frac{1}{s+3} x(0)}_{\text{transziens}} = 10$$

$\mathcal{L}\{u(t)\} = 1/s$

← vagy érdekes lehet nem beelyesíteni

$$x(t) = \underbrace{\mathcal{L}^{-1}\left\{\frac{7+0s}{s(s+3)}\right\}}_{\text{parciális törtekre bontás}} + 10 \underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}}_{e^{-3t} \varepsilon(t)}$$

$$\mathcal{L}\left\{\frac{1}{s+p}\right\} = e^{-pt} \varepsilon(t)$$

$$\frac{A}{s} + \frac{B}{s+3} = \frac{A(s+3) + Bs}{s(s+3)}$$

$$s^1: A + B = 0$$

$\Rightarrow$

$$A = 7/3$$

$$s^0: 3A = 7$$

$$B = -7/3$$

$$\frac{7}{3} \mathcal{L}^{-1}\left\{\frac{1}{s} - \frac{1}{s+3}\right\} = \frac{7}{3} (1 - e^{-3t}) \varepsilon(t)$$

$$x(t) = \left[ \frac{7}{3} (1 - e^{-3t}) + 10 e^{-3t} \right] \varepsilon(t) = \left[ \frac{7}{3} + \frac{23}{3} e^{-3t} \right] \varepsilon(t)$$

másik opció! **letakargató's módszer**

kifejtési tétel:

$$g_i = \lim_{s \rightarrow p_i} (s - p_i) \underbrace{\frac{\dots}{(s - p_1)(s - p_2) \dots (s - p_n)}}_{F(s)}$$

→  $p_i$ : pólusok

→ kimenet alakja:  $f(t) = (g_1 \cdot e^{p_1 t} + g_2 \cdot e^{p_2 t} + \dots + g_n \cdot e^{p_n t}) \varepsilon(t)$

→ FONTOS:

többszörös pólusnál  
NEM alkalmazható

$$\Rightarrow X(s) = \frac{7}{s(s+3)} + \frac{10}{s+3}$$

parciális törtre bontás helyett letakargatás módszer

$$F(s) = \frac{7}{s(s+3)} \rightarrow f(t) = (g_1 \cdot \underbrace{e^{0 \cdot t}}_1 + g_2 \cdot e^{-3t}) \varepsilon(t)$$

$p_1 = 0 \quad p_2 = -3$

$$g_1 = \lim_{s \rightarrow p_1} (s - p_1) F(s) = \lim_{s \rightarrow 0} s \cdot \frac{7}{s(s+3)} = \lim_{s \rightarrow 0} \frac{7}{s+3} = 7/3 = A$$

$$g_2 = \lim_{s \rightarrow p_2} (s - p_2) F(s) = \lim_{s \rightarrow -3} (s+3) \frac{7}{s(s+3)} = \lim_{s \rightarrow -3} \frac{7}{s} = -7/3 = B$$

$$f(t) = \frac{7}{3} (1 - e^{-3t}) \varepsilon(t)$$

↓

$$x(t) = \left[ \frac{7}{3} (1 - e^{-3t}) + 10 e^{-3t} \right] \varepsilon(t) = \left[ \frac{7}{3} + \frac{23}{3} e^{-3t} \right] \varepsilon(t)$$

(2)  $6\ddot{y}(t) + 2y(t) = u(t)$

$$y(0) = 5$$

$$\dot{y}(0) = -1$$

$$u(t) = 0$$

+ gyakorlati lehetőség:

$$u(t) = \varepsilon(t)$$

→ dx-es deriváltak?

$$\mathcal{L} \left\{ \frac{d^2}{dt^2} f(t) \right\} = \mathcal{L} \left\{ \underbrace{\left( \frac{d}{dt} \right) \left( \frac{d}{dt} \right) f(t)}_{g(t)} \right\} = \mathcal{L} \left\{ \frac{d}{dt} g(t) \right\} = sG(s) - g(0) =$$

$$g(t) = \left( \frac{d}{dt} \right) f(t)$$

$$= s \mathcal{L} \{ g(t) \} - g(0) = \boxed{s^2 F(s) - s f(0) - \dot{f}(0) = \mathcal{L} \left\{ \frac{d^2}{dt^2} f(t) \right\}}$$

$$= \mathcal{L} \left\{ \frac{d}{dt} f(t) \right\} = \dot{f}(0)$$

$$\rightarrow \underbrace{6(s^2 Y(s) - s y(0) - \dot{y}(0)) + 2Y(s)}_{\mathcal{L} \{ \ddot{y}(t) \}} = U(s)$$

$$\mathcal{L} \{ \ddot{y}(t) \}$$

$$Y(s) [6s^2 + 2] = \underbrace{U(s)}_0 + 6s \underbrace{y(0)}_5 + 6 \underbrace{\dot{y}(0)}_{-1} = 30s - 6$$

$$Y(s) = \frac{30s - 6}{6s^2 + 2} = \frac{5s - 1}{s^2 + 1/3}$$

$$\Rightarrow \text{pólusok: } s^2 + 1/3 = 0$$

$$s^2 = -1/3$$

$$s_{1,2} = \pm \frac{\sqrt{3}}{3} i$$

komplex konjugált gyökpár

↳ lehetséges megoldások:

- kiegészítés [ a) parciális törtekre bontás  
b) nem többszörös gyökök → letakargatás módszer  
c)  $\mathcal{L}^{-1}$  táblázat → itt:  $\sin$ ,  $\cos$  keresés

a) parciális törtekre bontás

$$Y(s) = \frac{5s - 1}{(s - 1/\sqrt{3}i)(s + 1/\sqrt{3}i)} = \frac{A}{s - 1/\sqrt{3}i} + \frac{B}{s + 1/\sqrt{3}i} = \frac{A(s + 1/\sqrt{3}i) + B(s - 1/\sqrt{3}i)}{(s - 1/\sqrt{3}i)(s + 1/\sqrt{3}i)}$$

$$s^1: A + B = 5 \longrightarrow A = 5 - B$$

$$s^0: A/\sqrt{3}i - B/\sqrt{3}i = -1$$

$$(A - B)i = -\sqrt{3}$$

$$(5 - B - B)i = -\sqrt{3}$$

$$5i + \sqrt{3} = 2Bi$$

$$B = \frac{5 + \sqrt{3}/i}{2} = \frac{5 - \sqrt{3}i}{2} \rightarrow A = 5 - \frac{5 - \sqrt{3}i}{2} = \frac{5 + \sqrt{3}i}{2}$$

$$\frac{1}{i} = \frac{i}{i^2} = -i$$

$$Y(s) = \frac{5 + \sqrt{3}i}{2} \cdot \frac{1}{s - 1/\sqrt{3}i} + \frac{5 - \sqrt{3}i}{2} \cdot \frac{1}{s + 1/\sqrt{3}i}$$

$\downarrow \mathcal{L}^{-1}$                        $\downarrow \mathcal{L}^{-1}$   
 $e^{i/\sqrt{3}t} \mathcal{E}(t)$                        $e^{-i/\sqrt{3}t} \mathcal{E}(t)$

$$y(t) = \left[ \frac{5 + \sqrt{3}i}{2} \cdot e^{i/\sqrt{3}t} + \frac{5 - \sqrt{3}i}{2} \cdot e^{-i/\sqrt{3}t} \right] \mathcal{E}(t)$$

b) letakargatás módszer

$$y(t) = (g_1 \cdot e^{p_1 t} + g_2 \cdot e^{p_2 t}) \mathcal{E}(t) = (g_1 \cdot e^{i/\sqrt{3}t} + g_2 \cdot e^{-i/\sqrt{3}t}) \mathcal{E}(t)$$

$$g_1 = \lim_{s \rightarrow p_1} (s - p_1) Y(s) = \lim_{s \rightarrow 1/\sqrt{3}i} = \cancel{(s - 1/\sqrt{3}i)} \frac{5s - 1}{\cancel{(s - 1/\sqrt{3}i)}(s + 1/\sqrt{3}i)} =$$

$$= \frac{5 \cdot 1/\sqrt{3}i - 1}{2/\sqrt{3}i} = \frac{5i - \sqrt{3}}{2i} = \frac{-5 - \sqrt{3}i}{-2} = \frac{5 + \sqrt{3}i}{2} = A$$

$$g_2 = \lim_{s \rightarrow p_2} (s - p_2) Y(s) = \lim_{s \rightarrow -1/\sqrt{3}i} = \cancel{(s + 1/\sqrt{3}i)} \frac{5s - 1}{\cancel{(s - 1/\sqrt{3}i)}(s + 1/\sqrt{3}i)} =$$

$$= \frac{-5 \cdot 1/\sqrt{3}i - 1}{-2/\sqrt{3}i} = \frac{-5i - \sqrt{3}}{-2i} = \frac{5 + \sqrt{3}i}{2} = B$$

$$\Rightarrow y(t) = \left[ \frac{5 + \sqrt{3}i}{2} \cdot e^{i/\sqrt{3}t} + \frac{5 - \sqrt{3}i}{2} \cdot e^{-i/\sqrt{3}t} \right] \varepsilon(t)$$

$$\hookrightarrow \sin(x) = \frac{e^{ix} - e^{-ix}}{2i} \quad \cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

átalakítás:

$$e^{i/\sqrt{3}t} \left( \frac{5}{2} + \frac{\sqrt{3}}{2}i \right) + e^{-i/\sqrt{3}t} \left( \frac{5}{2} - \frac{\sqrt{3}}{2}i \right) =$$

$$\underbrace{\frac{5}{2} - \frac{\sqrt{3}}{2i}} \quad \underbrace{\frac{5}{2} + \frac{\sqrt{3}}{2i}}$$

$$= \frac{5}{2} \left( e^{i/\sqrt{3}t} + e^{-i/\sqrt{3}t} \right) - \frac{\sqrt{3}}{2i} \left( e^{i/\sqrt{3}t} - e^{-i/\sqrt{3}t} \right) =$$

$$= 5 \cdot \underbrace{\frac{e^{i/\sqrt{3}t} + e^{-i/\sqrt{3}t}}{2}}_{\cos(1/\sqrt{3}t)} - \sqrt{3} \underbrace{\frac{e^{i/\sqrt{3}t} - e^{-i/\sqrt{3}t}}{2i}}_{\sin(1/\sqrt{3}t)t}$$

$$y(t) = \left[ 5 \cdot \cos(1/\sqrt{3}t) - \sqrt{3} \sin(1/\sqrt{3}t) \right] \varepsilon(t)$$

c) sin, cos keresés

$$Y(s) = \frac{5s - 1}{s^2 + 1/3} = 5 \frac{s}{s^2 + 1/3} - \frac{1}{s^2 + 1/3} = 5 \underbrace{\frac{s}{s^2 + (1/\sqrt{3})^2}}_{5 \cdot \frac{s}{s^2 + \omega^2}} - \underbrace{\frac{1 \cdot 1/\sqrt{3} \cdot \sqrt{3}}{s^2 + (1/\sqrt{3})^2}}_{\sqrt{3} \cdot \frac{\omega}{s^2 + \omega^2}}$$

$$y(t) = \left[ 5 \cdot \cos(1/\sqrt{3}t) - \sqrt{3} \sin(1/\sqrt{3}t) \right] \varepsilon(t)$$

a) rezgéstanosan

$$\ddot{y}(t) + \omega_n^2 y(t) = 0 \rightarrow \text{megoldásfüggvény:}$$

$$y(t) = C_1 \cdot \cos(\omega_n t) + C_2 \cdot \sin(\omega_n t)$$

$$\hookrightarrow \dot{y}(t) = -C_1 \cdot \omega_n \cdot \sin(\omega_n t) + C_2 \cdot \omega_n \cdot \cos(\omega_n t)$$

↓

$$y_0 = y(0) = \left[ C_1 \cdot \underbrace{\cos(0)}_1 + C_2 \cdot \underbrace{\sin(0)}_0 \right] \rightarrow y_0 = \underline{C_1}$$

$$\dot{y}_0 (= v_0) = -C_1 \cdot \omega_n \cdot \underbrace{\sin(0)}_0 + C_2 \cdot \omega_n \cdot \underbrace{\cos(0)}_1 \rightarrow \dot{y}_0 = C_2 \cdot \omega_n$$

$$C_2 = \underline{\dot{y}_0 / \omega_n}$$

$$\text{jelen esetben: } \omega_n = 1/\sqrt{3} \rightarrow C_1 = 5$$

$$C_2 = -\sqrt{3}$$

③  $\ddot{y} + 6\dot{y} + 8y = u$

$y(0) = 0$

$\dot{y}(0) = 0$

a) határozza meg a rendszer időállandóit és erősítését!

$$\mathcal{L} \Rightarrow s^2 Y(s) - s y(0) - \dot{y}(0) + 6[s Y(s) - y(0)] + 8 Y(s) = U(s)$$

$$Y(s) [s^2 + 6s + 8] = U(s) + \underbrace{[s + 6] y(0) + \dot{y}(0)}_{=0}$$

$$W(s) = \frac{Y(s)}{U(s)} = \frac{1}{s^2 + 6s + 8} = \frac{1}{\underbrace{(s+2)(s+4)}} = \frac{\underbrace{1/8}^*}{\underbrace{(1/2 s + 1)(1/4 s + 1)}}_{\substack{\text{gyöktényezőss} \\ \text{alak}} \quad \substack{\text{időállandós} \\ \text{alak}}}$$

$$\hookrightarrow \left. \begin{array}{l} T_1 = 1/2 = 0,5s \\ T_2 = 1/4 = 0,25s \end{array} \right\} \text{időállandók } (T_1 > T_2 > \dots > T_n)$$

$$K = 1/8 \quad ] \text{erősítés}$$

b)  $u(t) = \delta(t)$  - re mi a rendszer válasza?

$$\rightarrow U(s) = \mathcal{L}\{\delta(t)\} = 1$$

2. módszer: résztagozatós módszer

$$y(t) = (g_1 \cdot e^{-2t} + g_2 \cdot e^{-4t}) \varepsilon(t)$$

$$p_1 = -2 \quad p_2 = -4$$

$$g_1 = \lim_{s \rightarrow -2} (s+2) \frac{1}{(s+2)(s+4)} =$$

$$= \frac{1}{-2+4} = \frac{1}{2}$$

$$g_2 = \lim_{s \rightarrow -4} (s+4) \frac{1}{(s+2)(s+4)} =$$

$$= \frac{1}{-4+2} = -\frac{1}{2}$$

$\Downarrow$

$$y(t) = \frac{1}{2} (e^{-2t} - e^{-4t}) \varepsilon(t)$$

paralelis törtrekre bontás

$$\frac{A}{s+4} + \frac{B}{s+2} = \frac{A(s+2) + B(s+4)}{(s+4)(s+2)}$$

$$s^1: A + B = 0 \rightarrow A = -B$$

$$s^0: 2A + 4B = 1$$

$$-2B + 4B = 1$$

$$2B = 1$$

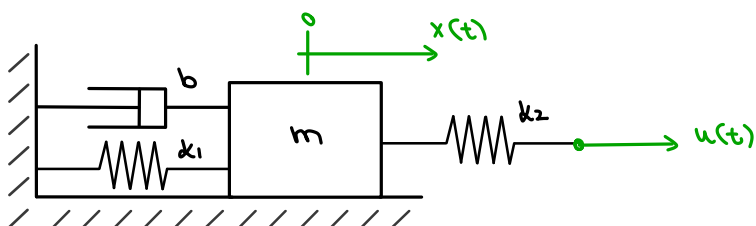
$$B = 1/2 \quad A = -1/2$$

$\Downarrow$

$$y(t) = \frac{1}{2} \left[ \mathcal{L}^{-1} \left\{ \frac{1}{s+2} \right\} - \mathcal{L}^{-1} \left\{ \frac{1}{s+4} \right\} \right] =$$

$$= \frac{1}{2} (e^{-2t} - e^{-4t}) \varepsilon(t)$$

(4)



$\rightarrow$  adatok:  $m = 10 \text{ kg}$

$$k_1 = 2250 \text{ N/m}$$

$$k_2 = 500 \text{ N/m}$$

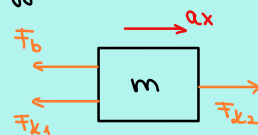
$$b = 600 \text{ Ns/m}$$

$$x(0) = 0 \text{ m}$$

$$v(0) = \dot{x}(0) = 0 \text{ m/s}$$

$$u(t) = \varepsilon(t)$$

megj.:



$$F_{k1} = k_1 x(t)$$

$$F_{k2} = (u(t) - x(t)) k_2$$

$$F_b = b \dot{x}(t)$$

$\rightarrow$  differenciáegyenlet:  $m\ddot{x}(t) + b\dot{x}(t) + (k_1 + k_2)x(t) = k_2 u(t)$

$$\xRightarrow{\mathcal{L}} m(s^2 X(s) - s x(0) - \dot{x}(0)) + b(s X(s) - x(0)) + (k_1 + k_2) X(s) = k_2 U(s)$$

$$X(s) [ms^2 + bs + k_1 + k_2] = k_2 U(s) + \underbrace{x(0)}_0 [ms + b] + \underbrace{\dot{x}(0)}_0 \cdot m$$

$$Y(s) = X(s) = \frac{k_2}{ms^2 + bs + k_1 + k_2} U(s) = \frac{k_2/m}{s^2 + b/ms + (k_1 + k_2)/m} U(s) =$$

pozíció  
a kimenet

$$= \frac{50}{s^2 + 60s + 275} U(s) = \frac{50}{(s+5)(s+55)} \cdot \frac{1}{s}$$

$U(s) = \mathcal{L}\{\varepsilon(t)\} = 1/s$

a) Letakargató's módszer

$$y(t) = (g_1 \cdot e^{0t} + g_2 \cdot e^{-5t} + g_3 \cdot e^{-55t}) \varepsilon(t)$$

$p_1 = 0$        $p_2 = -5$        $p_3 = -55$

$$g_1 = \lim_{s \rightarrow 0} s \cdot \frac{50}{(s+5)(s+55)} \cdot \frac{1}{s} = \frac{50}{5 \cdot 55} = \frac{10}{55} = \frac{2}{11}$$

$$g_2 = \lim_{s \rightarrow -5} (s+5) \cdot \frac{50}{(s+5)(s+55)} \cdot \frac{1}{s} = \frac{50}{(-5+55)(-5)} = -\frac{1}{5}$$

$$g_3 = \lim_{s \rightarrow -55} (s+55) \cdot \frac{50}{(s+5)(s+55)} \cdot \frac{1}{s} = \frac{50}{(-55+5)(-55)} = \frac{1}{55}$$

$\Downarrow$

$$y(t) = \left( \frac{2}{11} - \frac{1}{5} e^{-5t} + \frac{1}{55} e^{-55t} \right) \varepsilon(t)$$

b) parciális törtekre bontás  $\rightarrow$  feladat

$$Y(s) = \frac{A}{s} + \frac{B}{s+5} + \frac{C}{s+55} = \frac{\overbrace{A(s+5)(s+55) + Bs(s+55) + Cs(s+5)}^{s^2 + 60s + 275}}{s(s+5)(s+55)}$$

$$\begin{cases} s^2: A + B + C = 0 \\ s^1: 60A + 55B + 5C = 0 \\ s^0: 275A = 50 \end{cases}$$

$$A = 50/275 = 2/11 = g_1$$

$\downarrow$  visszahelyettesítés

$$B = -C - 2/11$$

$$60(2/11) + 55(-C - 2/11) + 5C = 0$$

$$120/11 - 110/11 = 55C - 5C$$

$$10/11 = 5C \rightarrow C = \frac{1}{55} = g_3$$

$\downarrow$



$$B = -1/55 - 2/11 = -11/55 = -1/5) = g_2$$

$$Y(s) = \frac{2}{11} \cdot \frac{1}{s} - \frac{1}{5} \cdot \frac{1}{s+5} + \frac{1}{55} \cdot \frac{1}{s+55}$$

$$y(t) = \frac{2}{11} \cdot \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \frac{1}{5} \cdot \mathcal{L}^{-1}\left\{\frac{1}{s+5}\right\} + \frac{1}{55} \cdot \mathcal{L}^{-1}\left\{\frac{1}{s+55}\right\} =$$

$$= \left( \frac{2}{11} - \frac{1}{5} e^{-5t} + \frac{1}{55} e^{-55t} \right) \varepsilon(t)$$

$$\textcircled{5} \ddot{x} + 3\dot{x} + 2x = 0$$

$$\begin{bmatrix} x(0) = 1 \\ \dot{x}(0) = 2 \end{bmatrix} \Rightarrow \underline{x}(0) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\Rightarrow \text{állapotter modelle: } \dot{\underline{x}} = \underline{A} \underline{x} + \underline{B} u$$

$$\begin{aligned} \text{legyen: } x_1 = x & \Rightarrow \ddot{x} = \ddot{x}_1 = \dot{x}_2 \rightarrow \dot{x}_1 = x_2 \\ x_2 = \dot{x} = \dot{x}_1 & \quad \quad \quad \dot{x}_2 = \ddot{x} = -3\dot{x} - 2x \\ & \quad \quad \quad = -3x_2 - 2x_1 \end{aligned}$$

mivel  $x_1 = s$   
 $x_2 = v = \dot{s}$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \dot{\underline{x}} = \underline{A} \underline{x} \quad (u=0)$$

→ tudjuk, hogy

$$\begin{aligned} \dot{\underline{x}} = \underline{A} \underline{x} & \xrightarrow{\mathcal{L}} s \underline{X}(s) - \underline{x}(0) = \underline{A} \underline{X}(s) \\ (s\mathbb{I} - \underline{A}) \underline{X}(s) &= \underline{x}(0) \\ \underline{X}(s) &= (s\mathbb{I} - \underline{A})^{-1} \underline{x}(0) \\ &\downarrow \mathcal{L}^{-1} \\ \underline{x}(t) &= \underline{x}(0) \cdot e^{\underline{A}t}, \quad t \geq 0 \end{aligned}$$

$$e^{\underline{A}t} = \mathcal{L}^{-1}\{(s\mathbb{I} - \underline{A})^{-1}\}$$

$$\begin{aligned} \rightarrow (s\mathbb{I} - \underline{A})^{-1} &= \left( \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \right)^{-1} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}^{-1} = \\ &= \frac{1}{s(s+3)+2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} = \end{aligned}$$

$\frac{1}{\det(\cdot)}$        $\text{adj}(\cdot)$

$$\underline{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$\det(\underline{A}) = ad - bc$   
determináns

$$\text{adj}(\underline{A}) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

adjungált

$$= \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

mit lehet tenni?

inverz Laplace elemenként

→ c'e'e?  $\underline{x}(t) = ?$

tudjuk, hogy  $\underline{x}(t) = \underbrace{e^{\underline{A}t}}_{\downarrow} \underline{x}(0)$

ez a kérdés,

legyen az alakja:

$$\begin{bmatrix} \psi_{11} & \psi_{12} \\ \psi_{21} & \psi_{22} \end{bmatrix} = \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \right\}$$

$$\psi_{11} = \mathcal{L}^{-1} \left\{ \frac{s+3}{(s+1)(s+2)} \right\} = g_1 \cdot e^{-t} + g_2 \cdot e^{-2t} = 2 \cdot e^{-t} - 1 \cdot e^{-2t}$$

$$g_1 = \lim_{s \rightarrow -1} \cancel{(s+1)} \frac{s+3}{\cancel{(s+1)}(s+2)} = \lim_{s \rightarrow -1} \frac{s+3}{s+2} = \frac{2}{1} = 2$$

$$g_2 = \lim_{s \rightarrow -2} \cancel{(s+2)} \frac{s+3}{(s+1)\cancel{(s+2)}} = \lim_{s \rightarrow -2} \frac{s+3}{s+1} = \frac{1}{-1} = -1$$

----- kiegészítés: többi  $\psi$  kiszámítása

$$\psi_{12} = \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)(s+2)} \right\} = g_1 \cdot e^{-t} + g_2 \cdot e^{-2t} = 1 \cdot e^{-t} - 1 \cdot e^{-2t}$$

$$g_1 = \lim_{s \rightarrow -1} \cancel{(s+1)} \frac{1}{\cancel{(s+1)}(s+2)} = \lim_{s \rightarrow -1} \frac{1}{s+2} = \frac{1}{1} = 1$$

$$g_2 = \lim_{s \rightarrow -2} \cancel{(s+2)} \frac{1}{(s+1)\cancel{(s+2)}} = \lim_{s \rightarrow -2} \frac{1}{s+1} = \frac{1}{-1} = -1$$

$$\psi_{21} = \mathcal{L}^{-1} \left\{ \frac{-2}{(s+1)(s+2)} \right\} = g_1 \cdot e^{-t} + g_2 \cdot e^{-2t} = 2 \cdot e^{-t} - 2 \cdot e^{-2t}$$

$$g_1 = \lim_{s \rightarrow -1} \cancel{(s+1)} \frac{-2}{\cancel{(s+1)}(s+2)} = \lim_{s \rightarrow -1} \frac{-2}{s+2} = \frac{-2}{1} = -2$$

$$g_2 = \lim_{s \rightarrow -2} \cancel{(s+2)} \frac{-2}{(s+1)\cancel{(s+2)}} = \lim_{s \rightarrow -2} \frac{-2}{s+1} = \frac{-2}{-1} = 2$$

$$\psi_{22} = \mathcal{L}^{-1} \left\{ \frac{s}{(s+1)(s+2)} \right\} = g_1 \cdot e^{-t} + g_2 \cdot e^{-2t} = -1 \cdot e^{-t} + 2 \cdot e^{-2t}$$

$$g_1 = \lim_{s \rightarrow -1} \cancel{(s+1)} \frac{s}{\cancel{(s+1)}(s+2)} = \lim_{s \rightarrow -1} \frac{s}{s+2} = \frac{-1}{1} = -1$$

$$g_2 = \lim_{s \rightarrow -2} \cancel{(s+2)} \frac{s}{(s+1)\cancel{(s+2)}} = \lim_{s \rightarrow -2} \frac{s}{s+1} = \frac{-2}{-1} = 2$$



$$\begin{aligned}\underline{x}(t) &= \begin{bmatrix} \psi_{11} & \psi_{12} \\ \psi_{21} & \psi_{22} \end{bmatrix} \begin{bmatrix} x(0) \\ \dot{x}(0) \end{bmatrix} = \begin{bmatrix} \psi_{11} x(0) + \psi_{12} \dot{x}(0) \\ \psi_{21} x(0) + \psi_{22} \dot{x}(0) \end{bmatrix} = \\ &= \begin{bmatrix} 2e^{-t} \cdot 1 - e^{-2t} \cdot 1 + e^{-t} \cdot 2 - e^{-2t} \cdot 2 \\ -2e^{-t} \cdot 1 + 2e^{-2t} \cdot 1 - e^{-t} \cdot 2 + 2e^{-2t} \cdot 2 \end{bmatrix} \\ &= \begin{bmatrix} 4e^{-t} - 3e^{-2t} \\ -4e^{-t} + 6e^{-2t} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\end{aligned}$$

ellenőrzés:  $x_1 = 4e^{-t} - 3e^{-2t}$

$x_2 = \dot{x}_1 = -4e^{-t} + 6e^{-2t} \quad \checkmark$

→ ha nem csinálnánk beelőre állapotát mondjuk:

$$\begin{aligned}\ddot{x} + 3\dot{x} + 2x &= 0 \quad \xRightarrow{\mathcal{L}} \quad (s^2 X(s) - sx(0) - \dot{x}(0)) + 3(sX(s) - x(0)) + 2X(s) = 0 \\ X(s) [s^2 + 3s + 2] &= \underbrace{x(0)}_1 [s+3] + \underbrace{\dot{x}(0)}_2\end{aligned}$$

$$X(s) = \frac{s+3+2}{s^2+3s+2} = \frac{s+5}{(s+2)(s+1)} \quad p_1 = -2 \quad p_2 = -1$$

$x(t) = g_1 \cdot e^{-2t} + g_2 \cdot e^{-t}$

$$g_1 = \lim_{s \rightarrow -2} \cancel{(s+2)} \frac{s+5}{\cancel{(s+2)}(s+1)} = \lim_{s \rightarrow -2} \frac{s+5}{s+1} = \frac{3}{-1} = -3$$

$$g_2 = \lim_{s \rightarrow -1} \cancel{(s+1)} \frac{s+5}{(s+2)\cancel{(s+1)}} = \lim_{s \rightarrow -1} \frac{s+5}{s+2} = \frac{4}{1} = 4$$

$x(t) = 4e^{-t} - 3e^{-2t}$

$\rightarrow \dot{x}(t) = -4e^{-t} + 6e^{-2t} \quad \text{ugyanaz}$



⊕ mi van ha  $u \neq 0$ ?

$$\begin{aligned}\dot{\underline{x}} &= \underline{A} \underline{x} + \underline{B} u \quad \xrightarrow{\mathcal{L}} \quad s \underline{X}(s) - \underline{x}(0) = \underline{A} \underline{X}(s) + \underline{B} u(s) \\ (s \underline{I} - \underline{A}) \underline{X}(s) &= \underline{x}(0) + \underline{B} u(s)\end{aligned}$$

$$\underline{X}(s) = (s \underline{I} - \underline{A})^{-1} \underline{x}(0) + (s \underline{I} - \underline{A})^{-1} \underline{B} u(s)$$

...